

HW 12 Complex Numbers

$$8. 22 + (-5 + 8i) + 10i = 17 + 18i$$

$$12. \sqrt{-5} \cdot \sqrt{-10} = \sqrt{5}i \cdot \sqrt{10}i = \sqrt{50}i^2 = -5\sqrt{2}$$

$$13. (\sqrt{-10})^2 = (\sqrt{10}i)^2 = 10i^2 = -10$$

$$16. (6-2i)(2-3i) = 12 - 18i - 4i + 6i^2 = 6 - 22i$$

$$25. \overline{5+3i} = 5-3i$$

$$(5+3i)(5-3i) = 25 - 9i^2 = 34$$

$$29. \overline{20i} = -20i$$

$$(20i)(-20i) = -400i^2 = 400$$

$$31. \overline{\sqrt{8}} = \sqrt{8}$$

$$(\sqrt{8})(\sqrt{8}) = 8$$

$$33. \frac{6}{i} \cdot \frac{-i}{-i} = \frac{-6i}{+1} = -6i$$

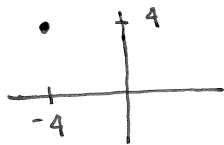
$$37. \frac{(2+i)}{(2-i)} \cdot \frac{(2+i)}{(2+i)} = \frac{4+4i+i^2}{4-i^2} = \frac{3}{5} + \frac{4}{5}i$$

$$55. -6i^3 + i^2 = 6i - 1 = -1 + 6i$$

$$58. (-i)^3 = -i^3 = i$$

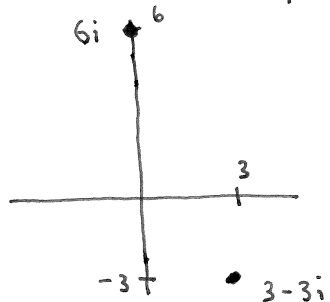
$$60. (\sqrt{-2})^6 = (\sqrt{2}i)^6 = 8i^6 = 8i^2 = -8$$

$$65. -4 + 4i$$



$$|-4 + 4i| = \sqrt{16 + 16} = 4\sqrt{2}$$

$$69. 3-3i$$



$$|3-3i| = \sqrt{9+9} = 3\sqrt{2} \quad \theta = -\frac{\pi}{4}$$

$$|6i| = 6$$

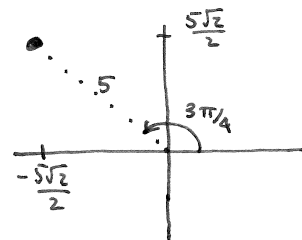
$$\theta = \frac{\pi}{2}$$

$$75. 6i$$

$$3-3i = 3\sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right) \right) = 3\sqrt{2} e^{-\frac{\pi}{4}i}$$

$$6i = 6 \left(\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right) \right) = 6 e^{\frac{\pi}{2}i}$$

$$78. 5 \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right) = 5 \left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \right) = -\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$$



$$83. 3 \left(\cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) \right) \cdot 4 \left(\cos\left(\frac{\pi}{6}\right) + i \sin\left(\frac{\pi}{6}\right) \right) \\ = 3e^{i\frac{\pi}{3}} \cdot 4e^{i\frac{\pi}{6}} = 12e^{i\frac{\pi}{2}} = 12 \left(\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right)$$

$$86. \frac{\cos\left(\frac{5\pi}{3}\right) + i \sin\left(\frac{5\pi}{3}\right)}{\cos\pi + i \sin\pi} = \frac{e^{i\frac{5\pi}{3}}}{e^{i\pi}} = e^{i\frac{2\pi}{3}} = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right)$$

$$87. (1+i)^5 = (\sqrt{2}e^{i\frac{\pi}{4}})^5 = (\sqrt{2})^5 e^{i\frac{5\pi}{4}} = 4\sqrt{2} \left(\cos\left(\frac{5\pi}{4}\right) + i \sin\left(\frac{5\pi}{4}\right) \right) \\ = -4 - 4i$$

$$r = \sqrt{2} \\ \theta = \frac{\pi}{4}$$

$$89. (-1+i)^{10} = (\sqrt{2}e^{i\frac{3\pi}{4}})^{10} = (\sqrt{2})^{10} e^{i\frac{15\pi}{2}} = 32 \left(\cos\left(\frac{15\pi}{2}\right) + i \sin\left(\frac{15\pi}{2}\right) \right) \\ = 32(0 - i) = -32i$$

$$r = \sqrt{2} \\ \theta = \frac{3\pi}{4}$$

$$94. \left[2 \left(\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right) \right]^8 = 2^8 (i)^8 = 2^8$$

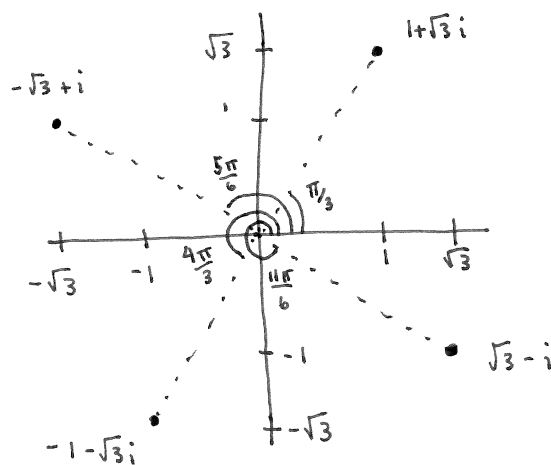
$$97. n=4 \\ r=16 \\ \theta = \frac{4\pi}{3} \\ \left(16 \left(\cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) \right) \right)^{1/4} = 2 \left(\cos\left(\frac{\frac{4\pi}{3} + 2\pi k}{4}\right) + i \sin\left(\frac{\frac{4\pi}{3} + 2\pi k}{4}\right) \right) \\ k=0,1,2,3 \\ = 2 \left(\cos\left(\frac{4\pi + 6\pi k}{12}\right) + i \sin\left(\frac{4\pi + 6\pi k}{12}\right) \right)$$

$$k=0: 2 \left(\cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) \right) = 2 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) = 1 + \sqrt{3}i$$

$$k=1: 2 \left(\cos\left(\frac{5\pi}{6}\right) + i \sin\left(\frac{5\pi}{6}\right) \right) = 2 \left(-\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = -\sqrt{3} + i$$

$$k=2: 2 \left(\cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) \right) = 2 \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2} \right) = -1 - \sqrt{3}i$$

$$k=3: 2 \left(\cos\left(\frac{11\pi}{6}\right) + i \sin\left(\frac{11\pi}{6}\right) \right) = 2 \left(\frac{\sqrt{3}}{2} - i \frac{1}{2} \right) = \sqrt{3} - i$$



$$101. x^4 - i = 0$$

$$x = (i)^{1/4} \rightarrow \begin{array}{l} n=4 \\ r=1 \\ \theta = \frac{\pi}{2} \end{array}$$

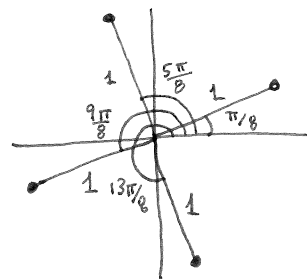
$$\begin{aligned} x &= r^{1/n} e^{i \left(\frac{\theta + 2\pi k}{n} \right)} \quad k=0, 1, 2, \dots, n-1 \\ &= 1^{1/4} e^{i \left(\frac{\pi/2 + 2\pi k}{4} \right)} \quad k=0, 1, 2, 3 \\ &= e^{i \left(\frac{\pi + 4\pi k}{8} \right)} \end{aligned}$$

$$k=0: x = e^{i \left(\frac{\pi}{8} \right)} \approx 0.924 + 0.383i$$

$$k=1: x = e^{i \frac{5\pi}{8}} \approx -0.383 + 0.924i$$

$$k=2: x = e^{i \frac{9\pi}{8}} \approx -0.924 - 0.383i$$

$$k=3: x = e^{i \frac{13\pi}{8}} \approx 0.383 - 0.924i$$



$$104. x^4 - 81 = 0$$

$$x^4 = (81)^{1/4} \rightarrow \begin{array}{l} n=4 \\ r=81 \\ \theta=0 \end{array}$$

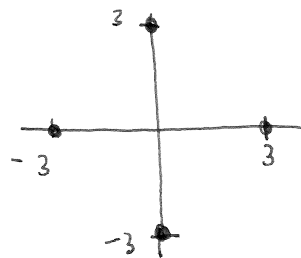
$$\begin{aligned} x &= (81)^{1/4} e^{i \left(\frac{2\pi k}{4} \right)} \\ &= 3 e^{i \frac{\pi k}{2}} \quad k=0, 1, 2, 3 \end{aligned}$$

$$k=0: x = 3e^0 = 3$$

$$k=1: x = 3e^{i \frac{\pi}{2}} = 3(0+i) = 3i$$

$$k=2: x = 3e^{i \pi} = 3(-1) = -3$$

$$k=3: x = 3e^{i \frac{3\pi}{2}} = 3(0-i) = -3i$$



$$107. x^3 - (1-i) = 0$$

$$x = (1-i)^{1/3} \rightarrow \begin{array}{l} n=3 \\ r=\sqrt{2} \\ \theta = -\pi/4 \end{array}$$

$$\begin{aligned} x &= (\sqrt{2})^{1/3} e^{i \left(\frac{-\pi/4 + 2\pi k}{3} \right)} \\ &= 2^{1/6} e^{i \left(\frac{-\pi + 8\pi k}{12} \right)} \quad k=0, 1, 2 \end{aligned}$$

$$k=0: x = 2^{1/6} e^{i \left(-\frac{\pi}{12} \right)} \approx 1.084 - 0.291i$$

$$k=1: x = 2^{1/6} e^{i \left(\frac{7\pi}{12} \right)} \approx -0.291 + 1.084i$$

$$k=2: x = 2^{1/6} e^{i \left(\frac{15\pi}{12} \right)} \approx -0.794 - 0.794i$$

